

SOUTH BEND, IN, USA

WMO: 725350

Lat: 41.707N

Lon: 86.316W

Elev: 236

StdP: 98.53

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14848

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.0	-15.0	-21.9	0.5	-17.6	-18.9	0.7	-14.3	11.8	-2.1	10.9	-1.9	4.3	230	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	32.3	23.3	30.8	22.4	29.4	21.8	25.1	30.0	24.1	28.6	23.2	27.4	5.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	18.9	28.1	22.6	17.8	26.8	21.7	16.8	25.9	77.5	29.9	73.3	28.8	69.8	27.4	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.2	8.3	DB	-22.2	34.8	3.9	1.7	-25.0	36.0	-27.3	36.9	-29.5	37.9	-32.4	39.1	(4)
(5)			WB	-22.5	26.8	3.9	1.2	-25.2	27.6	-27.5	28.4	-29.7	29.0	-32.5	29.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-4.1	-2.3	3.1	9.5	15.5	20.9	22.8	21.9	18.3	11.7	4.7	-0.9	(6)	
	DBStd	10.72	6.48	6.10	6.27	5.37	5.01	3.95	3.19	2.94	4.22	4.88	5.29	5.74	(7)	
	HDD10.0	1653	439	347	232	73	9	0	0	0	1	38	174	341	(8)	
	HDD18.3	3437	697	578	475	268	119	19	3	6	53	213	409	597	(9)	
	CDD10.0	1704	1	1	17	58	178	327	397	368	249	90	15	3	(10)	
	CDD18.3	447	0	0	1	4	30	96	142	116	51	7	0	0	(11)	
	CDH23.3	3809	0	0	9	41	291	854	1254	885	415	60	0	0	(12)	
	CDH26.7	1159	0	0	1	6	78	284	425	238	117	9	0	0	(13)	
(14) Wind	WSAvg	4.0	4.7	4.6	4.5	4.7	4.0	3.5	3.1	2.9	3.1	3.8	4.3	4.4	(14)	
(15) Precipitation	PrecAvg	992	62	55	69	92	92	105	96	98	90	85	77	72	(15)	
	PrecMax	1412	136	205	202	190	212	276	220	326	354	232	171	140	(16)	
	PrecMin	639	17	14	20	13	19	12	30	11	0	23	7	22	(17)	
	PrecStd	156	30	33	36	38	49	55	43	60	60	47	34	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	16.6	24.5	27.4	31.3	33.7	34.9	32.7	32.3	28.0	20.7	16.2	(19)	
		MCWB	12.1	12.3	16.9	17.4	20.5	23.2	25.5	24.6	22.2	19.8	13.7	14.1	(20)	
	2%	DB	10.6	12.5	20.0	24.1	28.8	31.7	32.4	30.9	29.9	25.0	17.4	12.7	(21)	
		MCWB	9.1	9.4	14.0	15.7	19.7	22.1	24.1	23.3	21.2	17.9	13.5	10.5	(22)	
	5%	DB	7.1	8.8	16.2	21.7	26.7	29.9	30.7	29.2	27.7	22.3	15.3	9.3	(23)	
		MCWB	5.4	6.3	11.8	14.3	18.7	21.6	23.0	22.4	20.4	16.7	11.9	7.2	(24)	
	10%	DB	4.2	5.8	12.9	19.0	24.4	28.1	29.0	27.8	25.6	19.9	12.9	6.5	(25)	
		MCWB	2.8	3.6	9.1	12.6	17.5	20.8	22.2	21.5	19.1	15.4	9.6	4.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	13.4	17.6	19.6	22.9	25.0	26.9	25.9	24.1	21.6	16.3	14.7	(27)	
		MCDB	13.9	15.3	24.5	25.3	28.3	31.1	32.8	31.1	29.6	26.1	18.7	16.2	(28)	
	2%	WB	9.2	9.7	15.0	17.1	21.2	23.9	25.5	24.7	22.8	19.5	14.4	10.9	(29)	
		MCDB	10.2	11.9	18.9	21.9	26.4	29.4	30.4	28.9	27.3	22.7	16.5	12.3	(30)	
	5%	WB	5.5	6.3	12.2	15.2	20.1	22.9	24.4	23.7	21.6	17.7	12.1	7.5	(31)	
		MCDB	6.7	8.7	15.5	20.2	24.9	27.9	29.0	27.4	25.4	21.1	14.7	9.1	(32)	
	10%	WB	2.9	3.6	9.2	13.5	18.6	21.8	23.4	22.7	20.5	15.8	10.1	4.8	(33)	
		MCDB	4.2	5.6	12.7	18.4	22.9	26.4	27.6	26.3	24.2	19.4	12.7	6.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	7.9	9.7	11.3	11.2	10.7	10.5	10.1	11.0	10.0	8.1	6.7	(35)	
		MCDBR	9.5	11.5	13.6	14.3	13.3	12.1	11.6	11.3	12.5	13.1	10.8	9.5	(36)	
	5% WB	MCWBR	8.2	9.2	9.1	8.2	6.8	5.3	5.2	5.2	5.6	7.2	7.8	8.3	(37)	
		MCWBR	9.0	10.7	12.1	12.7	11.7	10.4	10.2	9.4	10.5	11.1	9.5	8.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.301	0.325	0.345	0.395	0.424	0.430	0.442	0.438	0.414	0.355	0.318	0.299	(40)	
		taud	2.392	2.330	2.366	2.244	2.222	2.248	2.213	2.218	2.260	2.429	2.479	2.449	(41)	
	Ebn at Noon		846	877	901	874	855	848	832	823	816	834	823	819	(42)	
		Edh at Noon	80	100	109	133	139	136	140	135	121	91	74	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.46	2.28	3.54	4.66	5.33	6.08	6.14	5.33	4.37	2.83	1.74	1.22	(44)	
	RadStd		0.14	0.30	0.30	0.40	0.49	0.38	0.31	0.31	0.36	0.30	0.25	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page